

Do **not** write solutions on this page.

12. [Maximum mark: 21]

The equations of two lines, L_1 and L_2 , are given by:

$$L_1 : \mathbf{r}_1 = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \text{ where } s \in \mathbb{R}$$

$$L_2 : \frac{x+1}{2} = y-7 = \frac{z+5}{3}$$

- (a) Show that the position vector of the point of intersection, L_1 and L_2 is $\begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix}$. [3]

The plane Π contains the lines L_1 and L_2 .

- (b) (i) Write down the equation of Π , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ where $\lambda, \mu \in \mathbb{R}$.

- (ii) Given that $\mathbf{b} \times \mathbf{c} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$, show that the Cartesian equation of Π is $4x + y - 3z = 18$. [3]

The plane intersects the coordinate axes at $P(4.5, 0, 0)$, $Q(0, q, 0)$, and $R(0, 0, r)$.

- (c) Write down the value of

(i) q ;

(ii) r . [2]

- (d) Use a vector method to find the area of the triangle PQR. [5]

Another line, L_3 , is normal to Π and passes through the point of intersection of L_1 and L_2 .

- (e) Write down an equation for L_3 in the form $\mathbf{r}_3 = \mathbf{m} + \gamma\mathbf{n}$. [2]

- (f) Given that the point $S(-11, 5, 7)$ lies on the line L_3 , find γ . [2]

- (g) Hence, find the volume of pyramid PQRS. [4]

